

■ Conservation of mechanical energy

SYLLABUS CONTENT

- ▶ In the absence of frictional resistive forces, the total mechanical energy of a system is conserved.
- ▶ If mechanical energy is conserved, work is the amount of energy transformed between different forms of mechanical energy in a system, such as: kinetic energy, gravitational potential energy and elastic potential energy.

Applying the law of conservation of energy to mechanical systems:

$$\text{kinetic energy} + \text{gravitational potential energy} + \text{elastic potential energy} = \text{constant}$$

◆ **Conservative force** A force, the action of which conserves mechanical energy. There is no energy dissipation.

But this is only true if there are no frictional (resistive) forces acting. Such forces can be described as ‘non-conservative’. **Conservative forces** (such as gravitational forces, for example) conserve mechanical energy and do not involve the dissipation of energy. (More precisely: a *conservative force* is one for which the total work done in moving between two points is independent of the path taken.)

We have already explained that energy dissipation occurs in all mechanical systems to a greater or lesser extent, because of ever-present frictional / resistive forces (non-conservative forces). However, the equation above remains very useful for predicting ‘ideal’ outcomes, and for determining the amount of energy dissipated in other situations.

One of the most common examples of mechanical energy transfers is that between gravitational potential energy and kinetic energy. Assuming no resistive forces: change of gravitational potential energy = change of kinetic energy

$$mg\Delta h = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$$

If the mass starts or ends with zero velocity during the time being considered: $v^2(\text{or } u^2) = 2g\Delta h$.
leading to: $v(\text{or } u) = \sqrt{2g\Delta h}$.

This equation can be used to relate height with speed for any mass falling from rest, or any mass projected upwards to its highest point (assuming that gravity is the only force acting).

Note that the same equation can be obtained using the equation of motion (see Topic A.1):
 $v^2 = u^2 + 2as$, with u or v equal to zero, and using s instead of h .

WORKED EXAMPLE A3.8

A ball is thrown upwards with a speed of 23 m s^{-1} . Calculate the maximum height that it can reach.

Answer

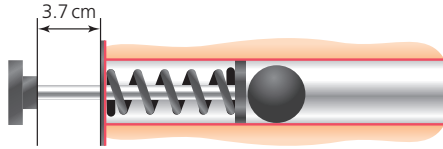
$$u^2 = 2g\Delta h$$

$$23^2 = 2 \times 9.8 \times \Delta h$$

$$\Delta h = 27 \text{ m}$$

WORKED EXAMPLE A3.9

A ball of mass 7.8 g was fired horizontally by a spring, as shown in Figure A3.30. The spring has a spring constant of 620 N m^{-1} and was pulled back 3.7 cm from its uncompressed length.



■ **Figure A3.30** Ball being fired horizontally by a spring

- Calculate how much elastic potential energy was stored in the spring. Assume that it obeys Hooke's law.
- What average force was used to compress the spring?
- Determine the maximum possible speed of the ball after the spring was released.
- Explain why the actual speed will be less than your answer to part c.

Answer

a $E_{\text{H}} = \frac{1}{2}k\Delta x^2 = 0.5 \times 620 \times 0.037^2 = 0.42 \text{ J}$

b $E_{\text{H}} = W = Fs$

$$0.42 = F \times 0.037$$

$$F = 11 \text{ N (This was half of the maximum force.)}$$

c $0.42 = \frac{1}{2}mv^2 = 0.5 \times 0.0078 \times v^2$

$$v = 10 \text{ m s}^{-1}$$

- d** All of the elastic potential energy in the spring was not transferred to the kinetic energy of the ball. For example, there was some friction with the sides of the tube, the spring did not stop moving after propelling the ball, there was some sound produced.

WORKED EXAMPLE A3.10

A box of mass 4.7 kg slid down a slope of vertical height 80 cm.

- Calculate the gravitational potential energy of the box at the top of the slope (compared to the bottom of the slope).
- Assuming the conservation of mechanical energy, what would the speed of the box be at the bottom of the slope?
- But the actual speed of the box was measured to be 2.2 m s^{-1} . Explain why the speed was less than the answer to part b.
- Determine how much energy was dissipated into the surroundings.
- In what form(s) was this dissipated energy?

Answer

a $\Delta E_{\text{p}} = mg\Delta h = 4.7 \times 9.8 \times 0.80 = 37 \text{ J}$

b $37 = \frac{1}{2}mv^2 = 0.5 \times 4.7 \times v^2$

$$v = 4.0 \text{ m s}^{-1}$$

Note that this answer (which assumes no energy dissipation) would be the same for any mass on any slope, or a vertical fall.

- c** Friction with the slope (and a little air resistance) acted in the opposite direction to motion.

d $37 - \frac{1}{2}mv^2 = 37 - (0.5 \times 4.7 \times 2.2^2) = 26 \text{ J}$

- e** Internal energy in the surfaces of the slope and box (which then spreads out as thermal energy).

22 What is the maximum speed with which a mass can hit the ground after being dropped from a height of 1.80 m?

23 The ball shown in Figure A3.31 was released from rest in position A. It accelerated down the slope and had its highest speed at the lowest point. It then moved up the slope on the other side, reaching its highest point at B.

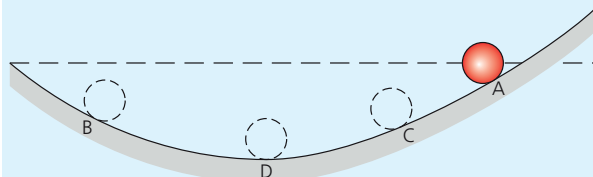


Figure A3.31 Ball rolling down slope

- Explain why B is lower than A.
 - Describe the motion of the ball after leaving position B, explaining the energy transfers until it finally comes to rest at D.
- 24** The large angle swing of a pendulum is interrupted by a peg as shown in Figure A3.32. Sketch a copy of the diagram and indicate the position of the pendulum after it has momentarily stopped moving on the right-hand side of the peg.

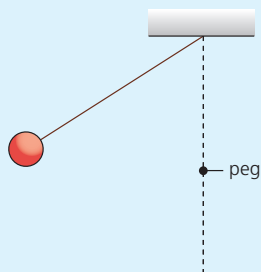


Figure A3.32 Pendulum

25 Figure A3.33 shows an experiment to measure the speed of a bullet which was fired into a block of wood. The bullet is embedded in the wood, so this was a totally inelastic collision. The block of wood and the bullet had kinetic energy and this was transferred to gravitational potential energy as they swung upwards.

- Outline what is meant by a totally inelastic collision.
- If the combined mass of the block and the bullet was 1.23 kg, determine their maximum gravitational potential energy.
- Use the law of conservation of energy to show that the initial velocity of the combined block and bullet was 1.5 m s^{-1} .

d If the bullet's mass was 15 g, use the law of conservation of momentum to determine its speed.

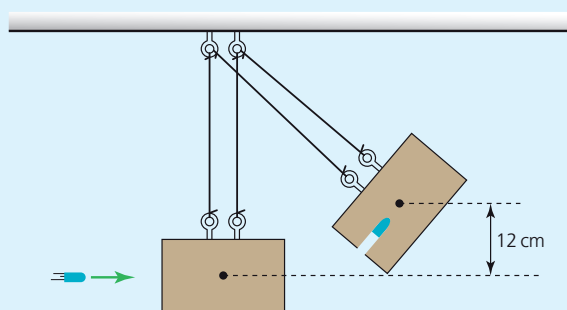


Figure A3.33 Experiment to measure the speed of a bullet

- A 10 g steel sphere moving to the left at 2.0 m s^{-1} collided with a similar sphere of mass 2.0 g moving in the opposite direction at 4.0 m s^{-1} . If after the collision the 10 g sphere remained stationary, determine what happened to the other sphere.
 - Calculate the total kinetic energy:
 - before the collision
 - after the collision.
 - Was mechanical energy conserved in this collision?
 - State the term we use to describe collisions like this.
- 27** In a laboratory experiment an 8.6 g wooden sphere moving at 0.39 m s^{-1} collided with a 5.7 g wooden sphere moving in the opposite direction with a speed of 0.72 m s^{-1} . After the collision they were both observed to move with a speed of 0.25 m s^{-1} , but in opposite directions.
- Show that these results are in good agreement with the law of conservation of momentum.
 - Calculate the total kinetic energy:
 - before the collision
 - after the collision.
 - Was kinetic energy conserved in this collision?
 - State the term we use to describe collisions like this.
- 28** A rubber band of mass 1.2 g was extended by 8.4 cm. The extension was proportional to the force and the band had a spring constant of 280 N m^{-1}
- If the force was released and the band fired vertically upwards, predict the maximum theoretical height that it could reach.
 - Explain why, in practice, the height will be a lot less than your answer to part a.

- 29** A long steel wire of mass 150 g was extended by 3.2 cm by a force which had increased slowly from 0 N to 240 N.
- Assuming that the extension was proportional to the force, how much elastic potential energy was stored in the wire?
 - The wire then snapped and the stored energy was transferred to the kinetic energy of the wire. Calculate an average value for the speed of the wire.
 - Discuss why breaking some metal wires can be dangerous.
- 30** ‘Crumple zones’ are a design feature of most vehicles (Figure A3.34). They are designed to compress and deform permanently if they are in a collision. Use the equation $E_s = \frac{1}{2}mv^2$ to help explain why a vehicle should not be too stiff and rigid.

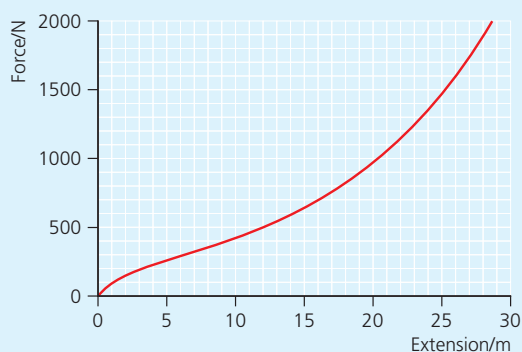


■ **Figure A3.34** The front of the car is deformed but the passenger compartment is intact

- 31** A bungee jumper (Figure A3.35) of mass 61 kg is moving at 23 m s^{-1} when the rubber bungee cord begins to become stretched.
- Calculate her kinetic energy at that moment.
 - Figure A3.36 shows how the extension of the cord varies with the applied force. State what quantity is represented by the area under this graph.
 - Describe the relationship between force and extension shown by this graph.
 - Use the graph to estimate how much the cord has extended by the time it has brought the jumper to a stop.



■ **Figure A3.35** Bungee jumping in Taupo, New Zealand



■ **Figure A3.36** Force–extension graph for a bungee cord

- 32** A pole-vaulter of mass 59.7 kg falls from a height of 4.76 m onto foam.
- Calculate the maximum possible kinetic energy on impact.
 - Will air resistance have had a significant effect in reducing the velocity of impact? Explain your answer.
 - If the foam deforms by 81 cm, estimate the average force exerted on the pole-vaulter.